

SEISMOGRAPH

A federated, privacy-preserving early-warning network for detecting silent behavioural drift in third-party LLM & agent APIs.

"Is it me, my prompt, or did the model silently change underneath me?"

Open Source · Apache-2.0

107 tests · CI-gated

OpenTelemetry-native

Differential Privacy

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Project

github.com/Tania-coder/SEISMOGRAPH
`pip install seismograph-probe`

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ABSTRACT

Teams that build on third-party large language models face a class of failure with no established defence: the model changes underneath them. Providers re-route, re-quantise, or re-tune deployed endpoints, and the resulting *semantic drift* produces no error code, no latency spike, and no uptime signal — yet it silently degrades downstream products. **SEISMOGRAPH** is a federated early-warning network that detects such drift by running lightweight, content-addressed *canary probes* against target APIs and correlating their distributional signals across independent observers. A strict privacy boundary guarantees that raw prompts and outputs never leave the probe perimeter: only hashed identifiers, distributional features, and differentially-private aggregates are transmitted. On a reproducible reconstruction of the publicly documented Anthropic Claude 3.5 Sonnet degradation of August–September 2025, a single SEISMOGRAPH probe raised its first drift alert **38 days before the official postmortem** — while the fault was still confined to ~0.8% of traffic and invisible to users. This paper describes the system, its detection method, its privacy guarantees, the validation result, and its honest current limitations.

1. The problem: silent semantic drift

Conventional observability watches infrastructure: latency, error rate, uptime. None of these move when a provider silently swaps the behaviour of a model behind a stable API endpoint. The output is still well-formed and the request still returns `200 OK` — it is simply *different*, and the difference propagates into every product built on top.

Such incidents are not hypothetical. In Q3 2025 Anthropic published a postmortem describing a load-balancer misconfiguration that misrouted a fraction of Claude 3.5 Sonnet traffic to an incompatible model configuration. The fault was introduced around **5 Aug 2025** at ~0.8% of traffic, escalated to ~16% by **29 Aug 2025**, and was publicly disclosed only on **17 Sep 2025**. For six weeks, downstream teams had no instrument that could distinguish “my prompt regressed” from “the model moved.” That diagnostic gap is the problem SEISMOGRAPH exists to close.

2. Approach: federated canary probes

SEISMOGRAPH deploys a small, fixed suite of **canary prompts** — a content-addressed, temperature-0 corpus capped at **200 prompts** — that is replayed against a target model on a schedule. Each run yields distributional features (e.g. JSON-validity rate, output-length distribution) rather than free text. Because the suite is immutable and hash-addressed, the same stimulus is compared against itself over time, so any movement in the response distribution is attributable to the model, not the test.

Crucially, no single observer can raise a public alarm. SEISMOGRAPH is **federated**: many organisations run probes independently, and a drift signal is promoted to a public alert only when *multiple distinct observers agree*. This correlation-first design separates a real provider-side change from one organisation’s local noise, prompt regression, or tampering.

≤ 200

CANARY
PROMPTS
@ TEMPERATURE
0

$< \$0.10$

TARGET COST
PER PROBE / DAY

≥ 2

OBSERVER
QUORUM
BEFORE PUBLIC
ALERT

$\epsilon = 2.0$

DP BUDGET
PER FLUSH
WINDOW

3. Detection method: two layers

Layer 1 — Per-observer change-point detection (CUSUM)

Each `(model_tuple, metric)` stream is monitored with a Page cumulative-sum (CUSUM) detector over standardised observations $z = (x - \mu_0) / \sigma_0$:

```
S+(n) = max(0, S+(n-1) + z(n) - k)    # detects positive shifts
S-(n) = max(0, S-(n-1) - z(n) - k)    # detects negative shifts
alert when S+(n) > h or S-(n) > h
```

Phase-0 parameters are $h = 5.0$ (threshold) and $k = 0.5$ (slack), the standard choices for unit-variance observations (Page 1954; $ARL_0 \approx 500$ at these settings). Each stream first accumulates **30 baseline observations** to estimate μ_0 and σ_0 before the detector activates; σ_0 is floored to avoid division by zero on constant series. The 30-sample baseline is itself a calibrated choice: an earlier 10-sample window underestimated σ_0 by 2.6× and produced false positives, a defect caught and corrected during validation.

Layer 2 — Cross-observer agreement (quorum gate)

Candidate alerts from Layer 1 are tagged with a pseudonymous observer identity and passed to the `AgreementScorer`. A candidate is promoted to a public alert only when the number of *distinct* observers reporting a change for the same model tuple meets the quorum (`QUORUM_MIN = 2`). Single-observer signals are retained as private fleet data and are never surfaced publicly. Replayed or duplicated signals are de-duplicated by observer identity within a scoring round, providing a first line of Sybil resistance.

4. Privacy by construction

The privacy boundary is the system's most important structural property. Raw prompts and raw outputs are destroyed at the probe's aggregation boundary and never appear in any transmitted object or downstream store. Three mechanisms enforce this:

- **Hashing.** Canary identifiers cross the wire only as non-reversible SHA-256 digests.
- **Differential privacy.** Numeric metrics carry calibrated Laplace noise at $\epsilon = 2.0$ per flush window (e.g. output-length sensitivity bounded by clamping to $[0, 8192]$). A `DPAccountant` enforces a rolling 24-hour budget via sequential composition, putting a probe to sleep before it can over-spend its privacy budget.

- **Pseudonymous federation.** Each probe holds an Ed25519 keypair; the engine identifies probes by public key and reputation, never by organisation identity. New keys start at zero reputation, so fabricated-signal injectors are down-weighted before they can influence a quorum.

This construction is what makes cross-organisation correlation viable at all: a competitor’s probe and yours can corroborate the same drift without either party revealing its prompts, its outputs, or even which model tuples it cares about beyond the noised aggregate.

5. Validation: the Anthropic Aug–Sep 2025 backtest

To test the core thesis — that distributional canaries surface silent drift early — we reconstructed the publicly documented incident as a reproducible simulation (`scripts/anthropic_backtest.py` , fixed seed). Daily canary observations were synthesised across three regimes matching the postmortem timeline, and a single Page-CUSUM detector was run over the `json_success_rate` stream.

Regime	Date range	json_success_rate	σ
Baseline	2025-07-01 → 08-04	0.990	0.006
Phase 1 (~0.8% misroute)	2025-08-05 → 08-28	0.982	0.006
Phase 2 (~16% misroute)	2025-08-29 → 09-17	0.840	0.015

The detector raised its first drift alert on **10 Aug 2025** (CUSUM $S_- = 7.28$ against threshold $h = 5.0$) — inside the subtle Phase 1 window, while the fault still affected only ~0.8% of traffic. That is **19 days before** the user-visible escalation and **38 days before** the official postmortem.

Date	Regime	json_rate	CUSUM S_-	
2025-08-04	Baseline	0.9881	0.000	
2025-08-05	Phase 1 (0.8%)	0.9855	0.598	
2025-08-07	Phase 1 (0.8%)	0.9786	3.309	
2025-08-09	Phase 1 (0.8%)	0.9816	4.889	
2025-08-10	Phase 1 (0.8%)	0.9777	7.278	◀ ALERT
2025-08-11	Phase 1 (0.8%)	0.9710	11.204	
2025-08-12	Phase 1 (0.8%)	0.9786	13.380	

Baseline statistics: $\mu_0 = 0.9903$, $\sigma_0 = 0.00437$ (`json_success_rate`). Detector: Page-CUSUM, $h = 5.0$, $k = 0.5$. Fully reproducible from a fixed seed; assertions C1–C6 pass.

6. Engineering & component status

SEISMOGRAPH is built as a working open-source system, not a paper design. The probe SDK is Python; the ingestion gateway is FastAPI; storage spans SQLite (validation) through Postgres and ClickHouse (scale); the public “model weather” dashboard is live. The codebase ships with **107 passing tests** behind a GitHub Actions CI gate (ruff + pytest) and an Apache-2.0 licence. We state component maturity honestly:

Component	Status	Notes
CUSUM change-point detector	live	Production-quality; used in the backtest and all validation.
Privacy / DP aggregation layer	live	Laplace noise, SHA-256 hashing, DP budget accountant.
Ingestion gateway + schema	live	Signed-batch validation; malformed batches rejected.
Cross-observer AgreementScorer	live	Quorum gate; single-org signals never promoted.
Webhooks & alerting	live	Fail-safe; auth tokens never logged; private-fleet path only.
Ed25519 signature verification	phase 2	Keypair + reputation designed; full verification in hardening.
OpenTelemetry span emission	phase 1	Interface defined (gen_ai.* / mcp.*); live wiring in progress.
Bayesian online detector	phase 1	Interface contract present; second-layer detector planned.

7. Limitations (stated honestly)

- **Synthetic validation.** The 38-day result is on a seeded reconstruction of the public incident, not live capture. Real lead time depends on probe cadence, traffic volume, and number of participating observers, and may be shorter or longer.
- **Single-observer simulation.** The backtest used one observer; a live public alert requires the ≥ 2 -observer quorum, which the simulation does not exercise end-to-end.
- **DP noise not in the sim.** Live probes add Laplace noise ($\epsilon=2.0$) that increases variance and may delay an alert by an estimated 1–3 days.
- **Calibration is provisional.** $h=5.0$, $k=0.5$ are Phase-0 defaults tuned on synthetic data; production deployment must recalibrate on real probe traffic.
- **Provider ToS.** Any production canary corpus must pass a per-provider terms-of-service review before deployment; Phase 0 uses mock execution only.

These limitations are the agenda for Phases 1–2, not caveats hidden in a footnote. The system’s governance model records every threshold decision and adversarial test as a labelled, auditable datum.

8. Roadmap

Phase	Milestone	Status
0 • Validation	Reproducible backtest proving detection lead-time vs a documented incident.	complete
1 • Public good	OSS probe + single-node ingestion + free public “model weather” dashboard.	core complete
2 • Network growth	DP hardening, OTel/MCP adapters, full Sybil resistance, ClickHouse, methodology paper.	in progress
3 • Enterprise	Multi-tenant, SSO/RBAC, SOC 2, in-VPC probe, SLAs, canary-gated rollback.	partial / planned

9. Reproducibility & availability

SEISMOGRAPH is open source under Apache-2.0. The probe is installable from PyPI (`pip install seismograph-probe`); the source, tests, and CI live at github.com/Tania-coder/SEISMOGRAPH; the public dashboard runs at seismograph-weather.onrender.com/dashboard. The validation backtest is reproducible from a fixed seed via `python3 scripts/anthropic_backtest.py`, with the full CUSUM trace recorded in `notebooks/anthropic_backtest_report.md` and the calibration record in `data/drift_labels/`. This release is permanently archived and citable at doi.org/10.5281/zenodo.21045518.

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